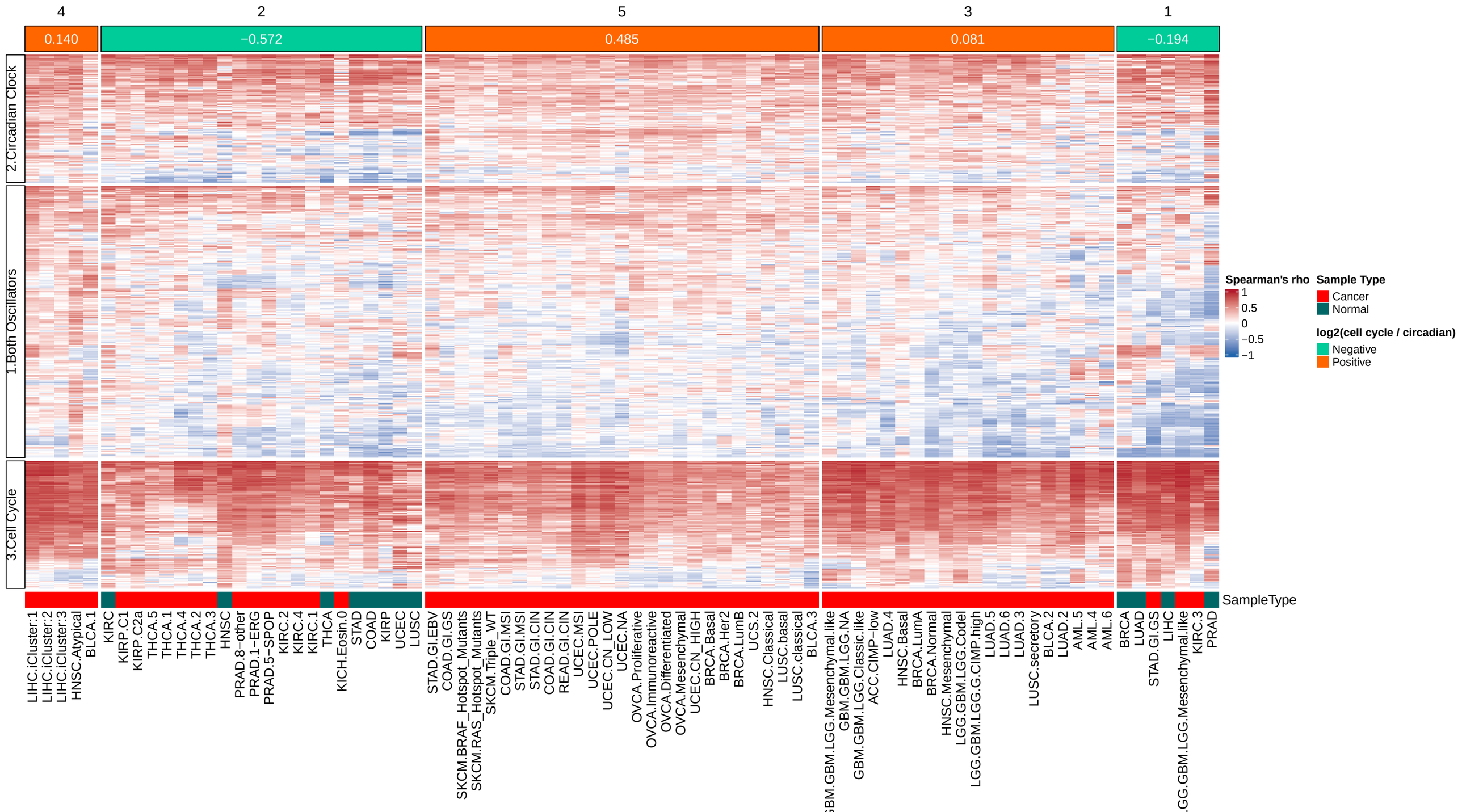
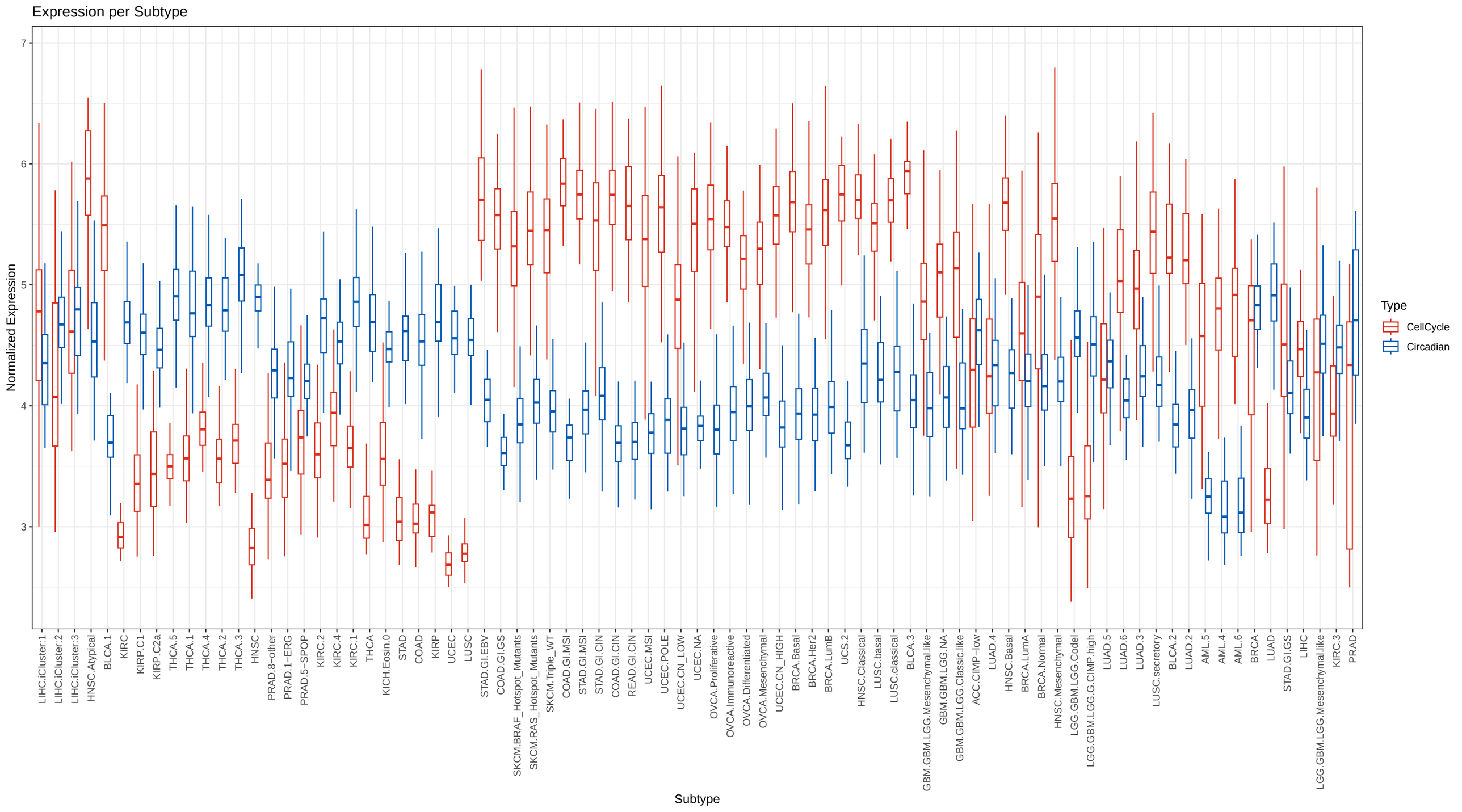
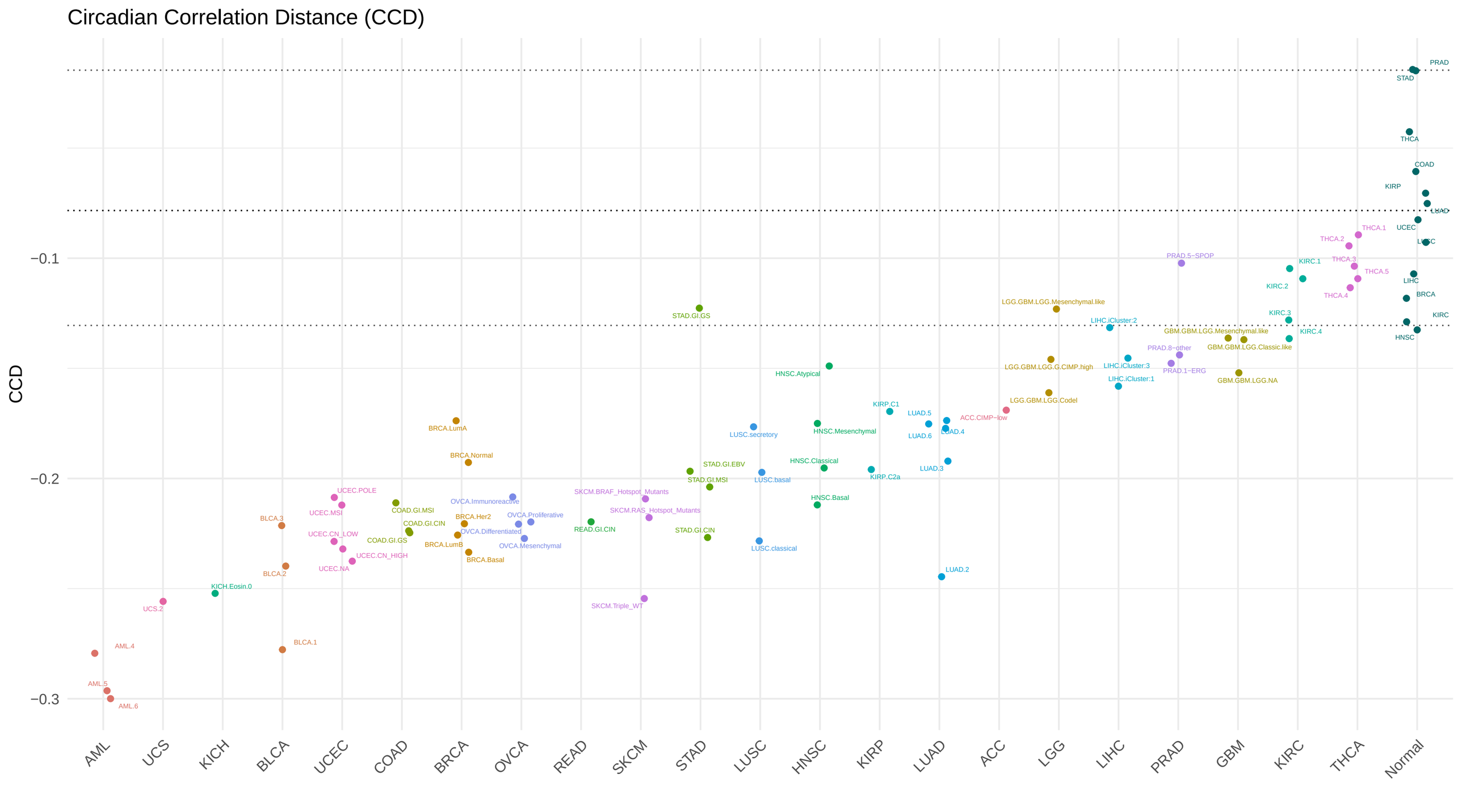
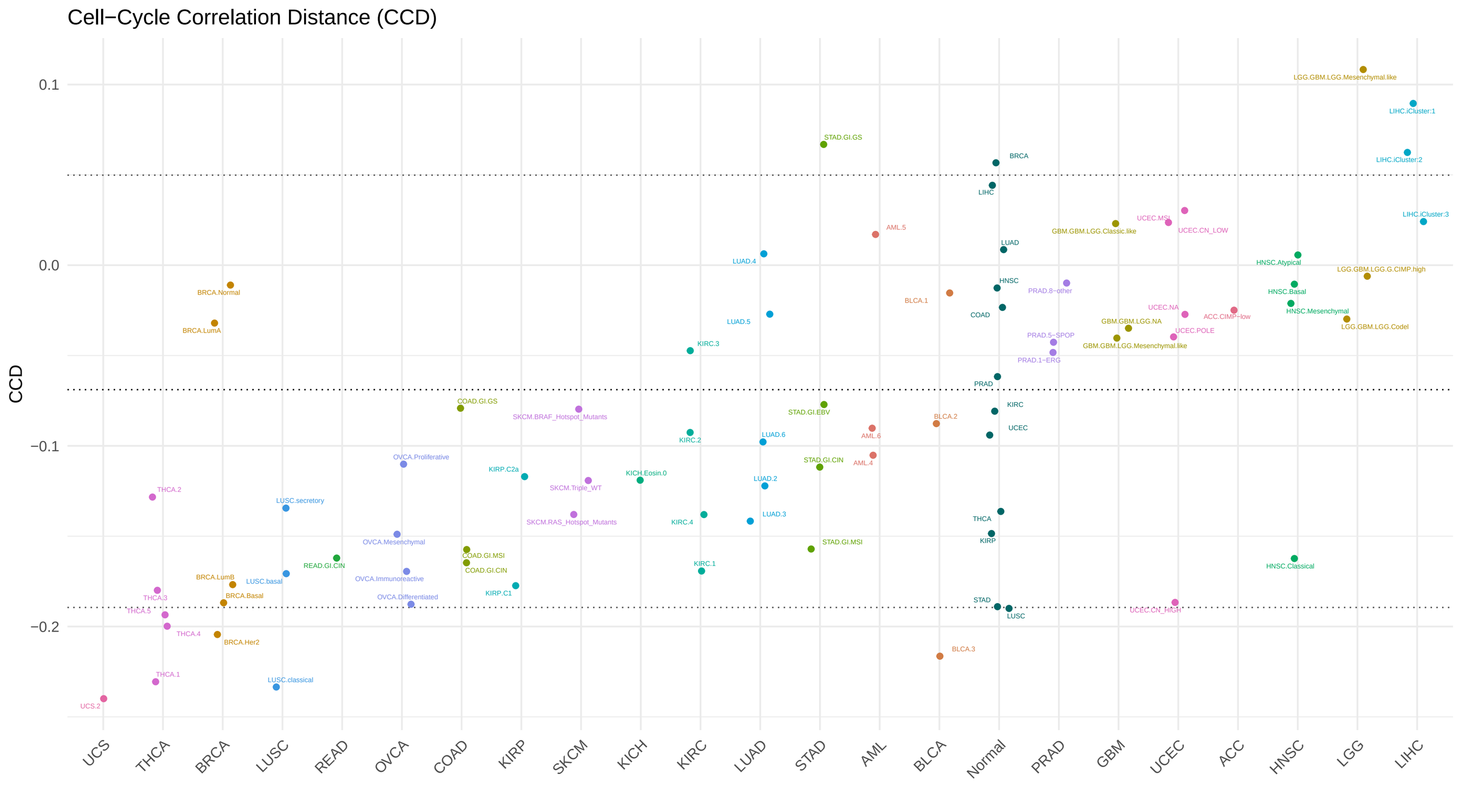
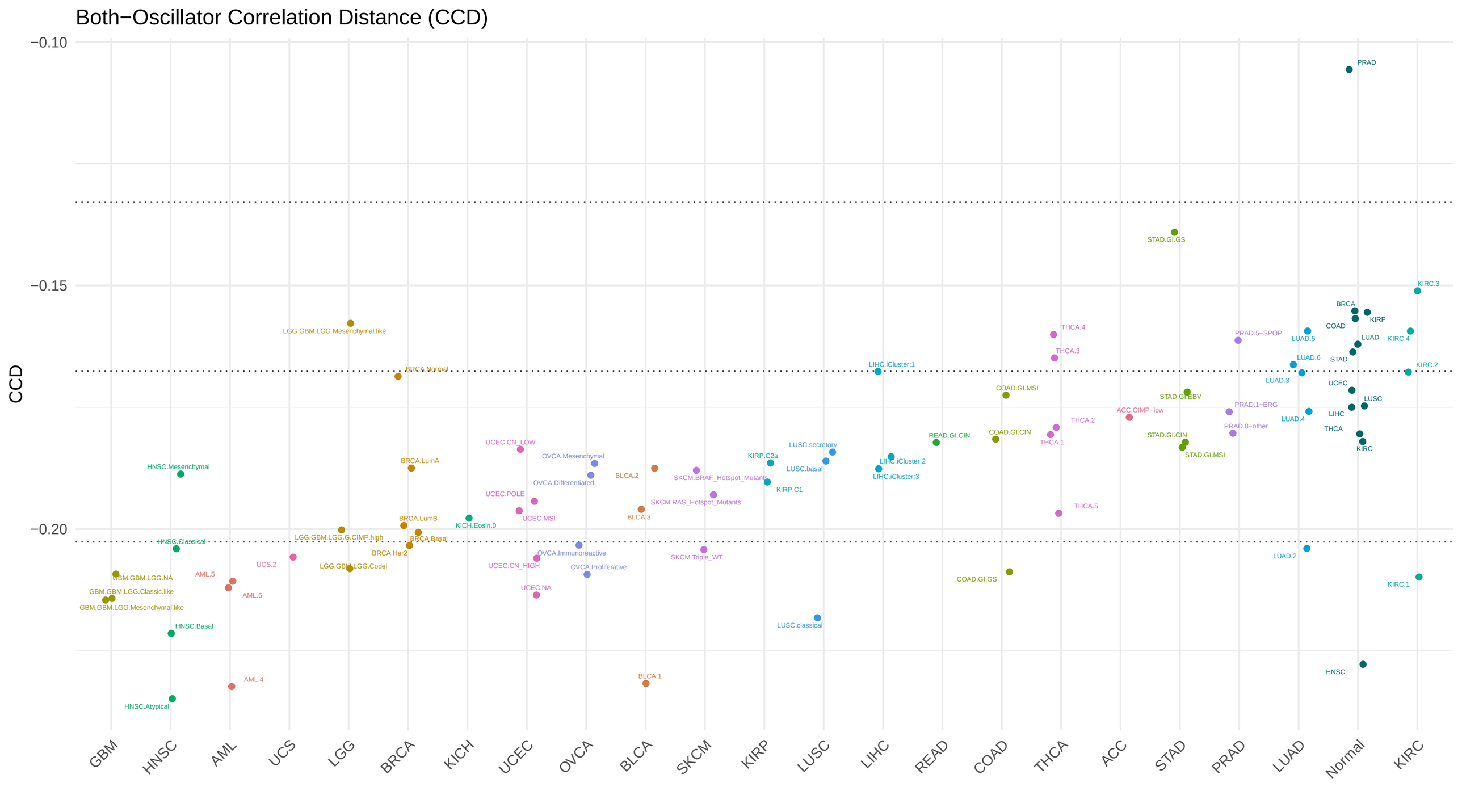
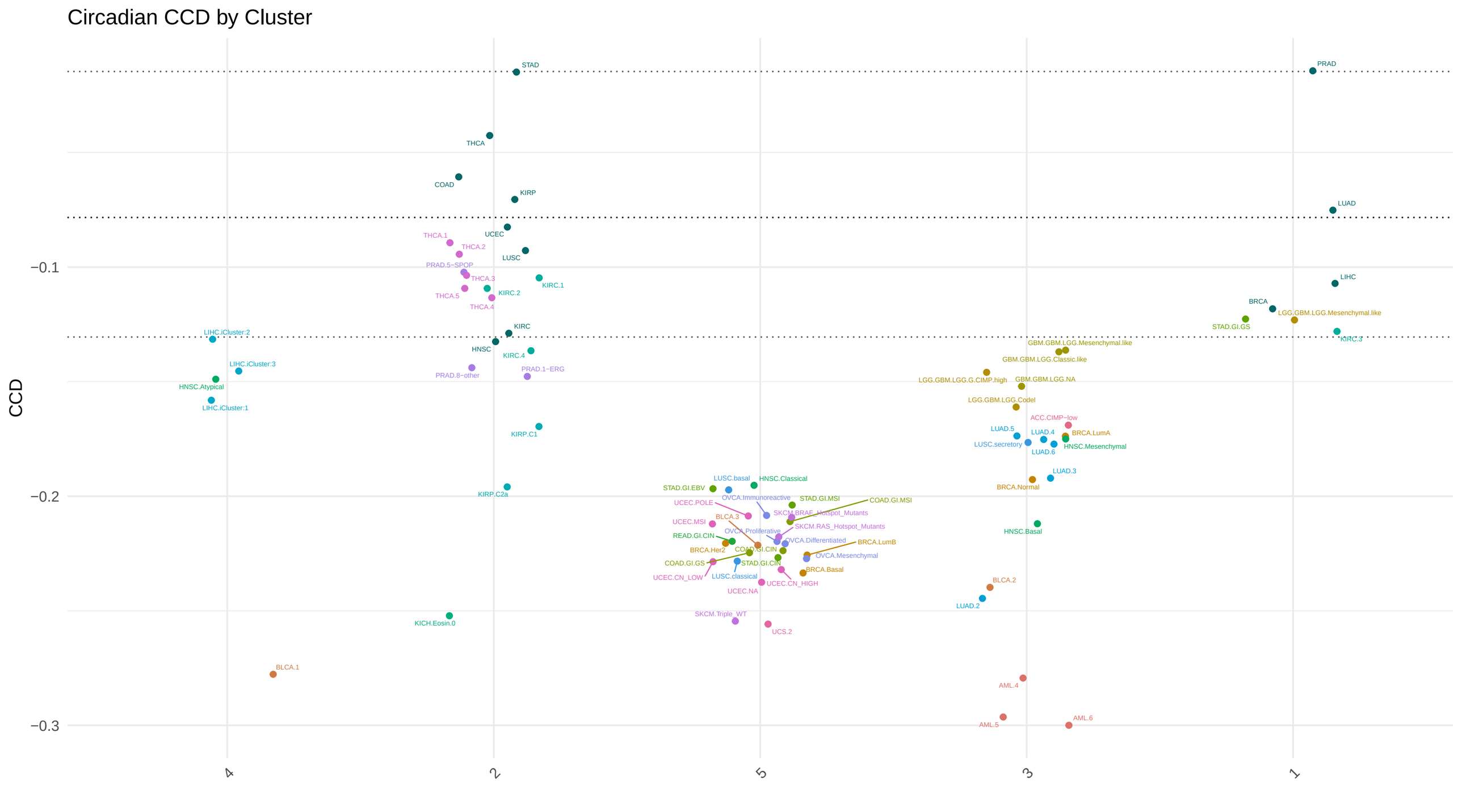

4
2
5
3
1
0.140
−0.572
0.485
0.081
−0.194
2.Circadian Clock
1.Both Oscillators
3.Cell Cycle
Spearman's rho
1
0.5
0
−0.5
−1
Sample Type
Cancer
Normal
log2(cell cycle / circadian)
Negative
Positive
SampleType
LIHC.iCluster:1
LIHC.iCluster:2
LIHC.iCluster:3
HNSC.Atypical
BLCA.1
KIRC
KIRP.C1
KIRP.C2a
THCA.5
THCA.1
THCA.4
THCA.2
THCA.3
HNSC
PRAD.8−other
PRAD.1−ERG
PRAD.5−SPOP
KIRC.2
KIRC.4
KIRC.1
THCA
KICH.Eosin.0
STAD
COAD
KIRP
UCEC
LUSC
STAD.GI.EBV
COAD.GI.GS
SKCM.BRAF_Hotspot_Mutants
SKCM.RAS_Hotspot_Mutants
SKCM.Triple_WT
COAD.GI.MSI
STAD.GI.MSI
STAD.GI.CIN
COAD.GI.CIN
READ.GI.CIN
UCEC.MSI
UCEC.POLE
UCEC.CN_LOW
UCEC.NA
OVCA.Proliferative
OVCA.Immunoreactive
OVCA.Differentiated
OVCA.Mesenchymal
UCEC.CN_HIGH
BRCA.Basal
BRCA.Her2
BRCA.LumB
UCS.2
HNSC.Classical
LUSC.basal
LUSC.classical
BLCA.3
GBM.GBM.LGG.Mesenchymal.like
GBM.GBM.LGG.NA
GBM.GBM.LGG.Classic.like
ACC.CIMP−low
LUAD.4
HNSC.Basal
BRCA.LumA
BRCA.Normal
HNSC.Mesenchymal
LGG.GBM.LGG.Codel
LGG.GBM.LGG.G.CIMP.high
LUAD.5
LUAD.6
LUAD.3
LUSC.secretory
BLCA.2
LUAD.2
AML.5
AML.4
AML.6
BRCA
LUAD
STAD.GI.GS
LIHC
LGG.GBM.LGG.Mesenchymal.like
KIRC.3
PRAD

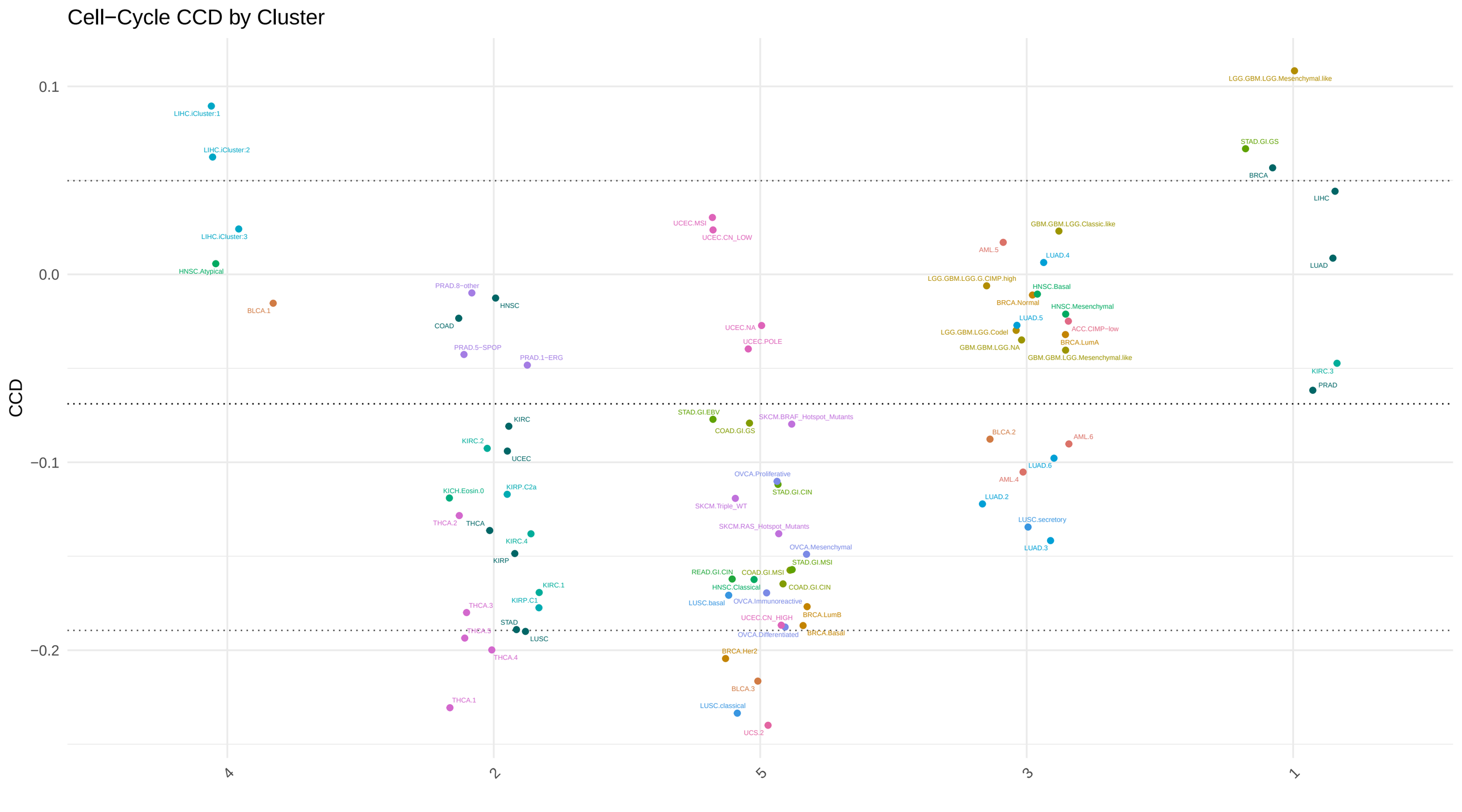

Expression per Subtype
Normalized Expression
7
6
5
4
3
LIHC.iCluster:1
LIHC.iCluster:2
LIHC.iCluster:3
HNSC.Atypical
BLCA.1
KIRC
KIRP.C1
KIRP.C2a
THCA.5
THCA.1
THCA.4
THCA.2
THCA.3
HNSC
PRAD.8−other
PRAD.1−ERG
PRAD.5−SPOP
KIRC.2
KIRC.4
KIRC.1
THCA
KICH.Eosin.0
STAD
COAD
KIRP
UCEC
LUSC
STAD.GI.EBV
COAD.GI.GS
SKCM.BRAF_Hotspot_Mutants
SKCM.RAS_Hotspot_Mutants
SKCM.Triple_WT
COAD.GI.MSI
STAD.GI.MSI
STAD.GI.CIN
COAD.GI.CIN
READ.GI.CIN
UCEC.MSI
UCEC.POLE
UCEC.CN_LOW
UCEC.NA
OVCA.Proliferative
OVCA.Immunoreactive
OVCA.Differentiated
OVCA.Mesenchymal
UCEC.CN_HIGH
BRCA.Basal
BRCA.Her2
BRCA.LumB
UCS.2
HNSC.Classical
LUSC.basal
LUSC.classical
BLCA.3
GBM.GBM.LGG.Mesenchymal.like
GBM.GBM.LGG.NA
GBM.GBM.LGG.Classic.like
ACC.CIMP−low
LUAD.4
HNSC.Basal
BRCA.LumA
BRCA.Normal
HNSC.Mesenchymal
LGG.GBM.LGG.Codel
LGG.GBM.LGG.G.CIMP.high
LUAD.5
LUAD.6
LUAD.3
LUSC.secretory
BLCA.2
LUAD.2
AML.5
AML.4
AML.6
BRCA
LUAD
STAD.GI.GS
LIHC
LGG.GBM.LGG.Mesenchymal.like
KIRC.3
PRAD
Subtype
Type
CellCycle
Circadian

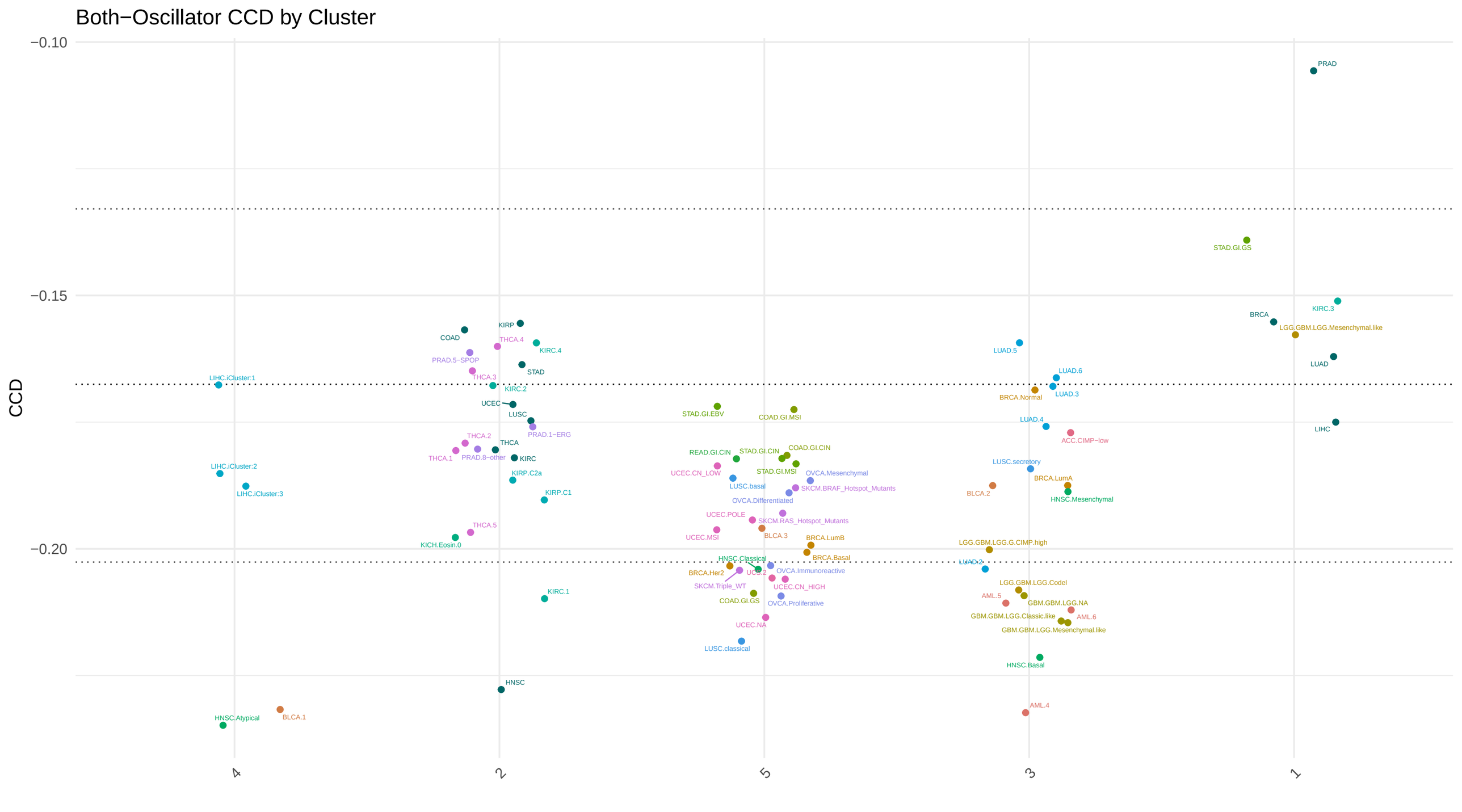

Circadian Correlation Distance (CCD)
CCD
−0.1
−0.2
−0.3
AML
UCS
KICH
BLCA
UCEC
COAD
BRCA
OVCA
READ
SKCM
STAD
LUSC
HNSC
KIRP
LUAD
ACC
LGG
LIHC
PRAD
GBM
KIRC
THCA
Normal
PRAD
STAD
THCA
COAD
KIRP
LUAD
UCEC
LUSC
PRAD.5−SPOP
THCA.1
THCA.2
KIRC.1
THCA.3
THCA.5
LIHC
KIRC.2
THCA.4
BRCA
KIRC.3
KIRC
KIRC.4
HNSC
LGG.GBM.LGG.Mesenchymal.like
LIHC.iCluster:2
GBM.GBM.LGG.Mesenchymal.like
GBM.GBM.LGG.Classic.like
PRAD.8−other
LGG.GBM.LGG.G.CIMP.high
LIHC.iCluster:3
PRAD.1−ERG
LIHC.iCluster:1
GBM.GBM.LGG.NA
LGG.GBM.LGG.Codel
ACC.CIMP−low
KIRP.C1
LUAD.5
LUAD.4
LUAD.6
LUAD.3
LUAD.2
HNSC.Atypical
HNSC.Mesenchymal
HNSC.Classical
HNSC.Basal
KIRP.C2a
LUSC.secretory
LUSC.basal
LUSC.classical
STAD.GI.GS
STAD.GI.EBV
STAD.GI.MSI
STAD.GI.CIN
BRCA.LumA
BRCA.Normal
BRCA.Her2
BRCA.LumB
BRCA.Basal
OVCA.Immunoreactive
OVCA.Proliferative
OVCA.Differentiated
OVCA.Mesenchymal
READ.GI.CIN
SKCM.BRAF_Hotspot_Mutants
SKCM.RAS_Hotspot_Mutants
SKCM.Triple_WT
UCEC.POLE
UCEC.MSI
UCEC.CN_LOW
UCEC.CN_HIGH
UCEC.NA
COAD.GI.MSI
COAD.GI.CIN
COAD.GI.GS
BLCA.3
BLCA.2
BLCA.1
KICH.Eosin.0
UCS.2
AML.4
AML.5
AML.6

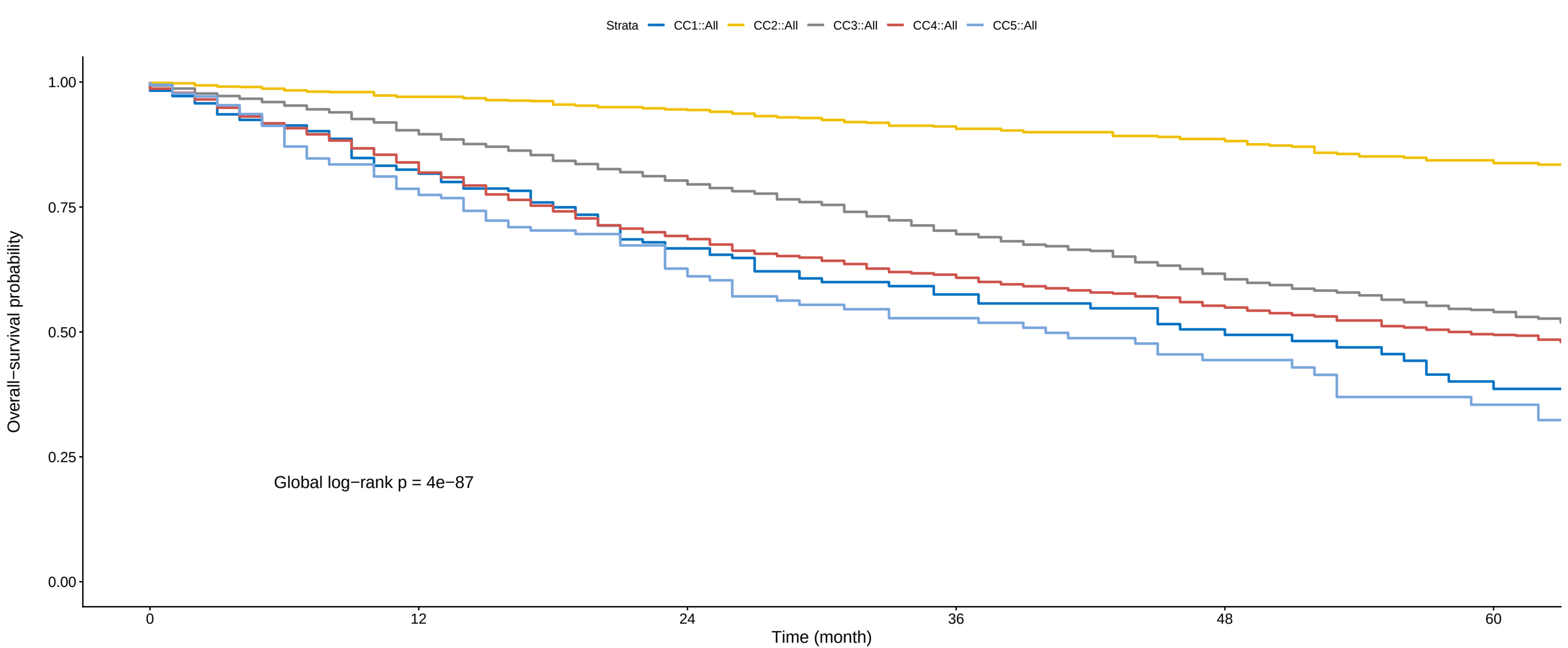

Cell−Cycle Correlation Distance (CCD)
CCD
0.1
0.0
−0.1
−0.2
UCS
THCA
BRCA
LUSC
READ
OVCA
COAD
KIRP
SKCM
KICH
KIRC
LUAD
STAD
AML
BLCA
Normal
PRAD
GBM
UCEC
ACC
HNSC
LGG
LIHC
LGG.GBM.LGG.Mesenchymal.like
LIHC.iCluster:1
LIHC.iCluster:2
LIHC.iCluster:3
STAD.GI.GS
BRCA
LIHC
AML.5
UCEC.MSI
UCEC.CN_LOW
GBM.GBM.LGG.Classic.like
LUAD
LUAD.4
HNSC.Atypical
LGG.GBM.LGG.G.CIMP.high
HNSC
PRAD.8−other
HNSC.Basal
BLCA.1
UCEC.NA
ACC.CIMP−low
HNSC.Mesenchymal
COAD
GBM.GBM.LGG.NA
LGG.GBM.LGG.Codel
LUAD.5
UCEC.POLE
PRAD.5−SPOP
KIRC.3
GBM.GBM.LGG.Mesenchymal.like
PRAD.1−ERG
PRAD
BRCA.Normal
BRCA.LumA
COAD.GI.GS
SKCM.BRAF_Hotspot_Mutants
STAD.GI.EBV
KIRC
BLCA.2
UCEC
KIRC.2
LUAD.6
AML.6
OVCA.Proliferative
STAD.GI.CIN
AML.4
KIRP.C2a
KICH.Eosin.0
LUAD.2
SKCM.Triple_WT
THCA.2
LUSC.secretory
LUAD.3
SKCM.RAS_Hotspot_Mutants
KIRC.4
THCA
OVCA.Mesenchymal
KIRP
STAD.GI.MSI
COAD.GI.MSI
READ.GI.CIN
KIRC.1
HNSC.Classical
BRCA.LumB
LUSC.basal
COAD.GI.CIN
OVCA.Immunoreactive
THCA.3
KIRP.C1
BRCA.Basal
OVCA.Differentiated
STAD
UCEC.CN_HIGH
THCA.5
LUSC
THCA.4
BRCA.Her2
BLCA.3
THCA.1
LUSC.classical
UCS.2


Both−Oscillator Correlation Distance (CCD)
CCD
−0.10
−0.15
−0.20
GBM
HNSC
AML
UCS
LGG
BRCA
KICH
UCEC
OVCA
BLCA
SKCM
KIRP
LUSC
LIHC
READ
COAD
THCA
ACC
STAD
PRAD
LUAD
Normal
KIRC
GBM.GBM.LGG.NA
GBM.GBM.LGG.Classic.like
GBM.GBM.LGG.Mesenchymal.like
HNSC.Mesenchymal
HNSC.Classical
HNSC.Basal
HNSC.Atypical
AML.5
AML.6
AML.4
UCS.2
LGG.GBM.LGG.Mesenchymal.like
LGG.GBM.LGG.Codel
LGG.GBM.LGG.G.CIMP.high
BRCA.Normal
BRCA.LumA
BRCA.LumB
BRCA.Basal
BRCA.Her2
KICH.Eosin.0
UCEC.CN_LOW
UCEC.POLE
UCEC.MSI
UCEC.CN_HIGH
UCEC.NA
OVCA.Mesenchymal
OVCA.Differentiated
OVCA.Immunoreactive
OVCA.Proliferative
BLCA.2
BLCA.3
BLCA.1
SKCM.BRAF_Hotspot_Mutants
SKCM.RAS_Hotspot_Mutants
SKCM.Triple_WT
KIRP.C2a
KIRP.C1
LUSC.secretory
LUSC.basal
LUSC.classical
LIHC.iCluster:1
LIHC.iCluster:2
LIHC.iCluster:3
READ.GI.CIN
COAD.GI.MSI
COAD.GI.CIN
COAD.GI.GS
THCA.4
THCA.3
THCA.2
THCA.1
THCA.5
ACC.CIMP−low
STAD.GI.GS
STAD.GI.EBV
STAD.GI.CIN
STAD.GI.MSI
PRAD
PRAD.5−SPOP
PRAD.1−ERG
PRAD.8−other
LUAD.5
LUAD.6
LUAD.3
LUAD.4
LUAD.2
BRCA
KIRP
COAD
LUAD
KIRC.4
STAD
KIRC.3
KIRC.2
UCEC
LUSC
LIHC
THCA
KIRC
KIRC.1
HNSC


Circadian CCD by Cluster
CCD
−0.1
−0.2
−0.3
4
2
5
3
1
STAD
PRAD
THCA
COAD
KIRP
LUAD
UCEC
THCA.1
THCA.2
LUSC
PRAD.5−SPOP
THCA.3
KIRC.1
LIHC
THCA.5
KIRC.2
BRCA
THCA.4
LGG.GBM.LGG.Mesenchymal.like
STAD.GI.GS
KIRC
KIRC.3
LIHC.iCluster:2
HNSC
GBM.GBM.LGG.Mesenchymal.like
KIRC.4
GBM.GBM.LGG.Classic.like
LIHC.iCluster:3
PRAD.8−other
PRAD.1−ERG
LGG.GBM.LGG.G.CIMP.high
GBM.GBM.LGG.NA
HNSC.Atypical
LGG.GBM.LGG.Codel
LIHC.iCluster:1
ACC.CIMP−low
LUAD.5
LUAD.4
BRCA.LumA
KIRP.C1
LUSC.secretory
HNSC.Mesenchymal
LUAD.6
LUAD.3
LUSC.basal
HNSC.Classical
BRCA.Normal
STAD.GI.EBV
KIRP.C2a
OVCA.Immunoreactive
STAD.GI.MSI
COAD.GI.MSI
UCEC.POLE
SKCM.BRAF_Hotspot_Mutants
BLCA.3
UCEC.MSI
HNSC.Basal
SKCM.RAS_Hotspot_Mutants
OVCA.Proliferative
READ.GI.CIN
OVCA.Differentiated
BRCA.LumB
BRCA.Her2
COAD.GI.CIN
OVCA.Mesenchymal
COAD.GI.GS
STAD.GI.CIN
BRCA.Basal
UCEC.CN_LOW
LUSC.classical
BLCA.2
UCEC.CN_HIGH
UCEC.NA
LUAD.2
SKCM.Triple_WT
KICH.Eosin.0
UCS.2
BLCA.1
AML.4
AML.5
AML.6

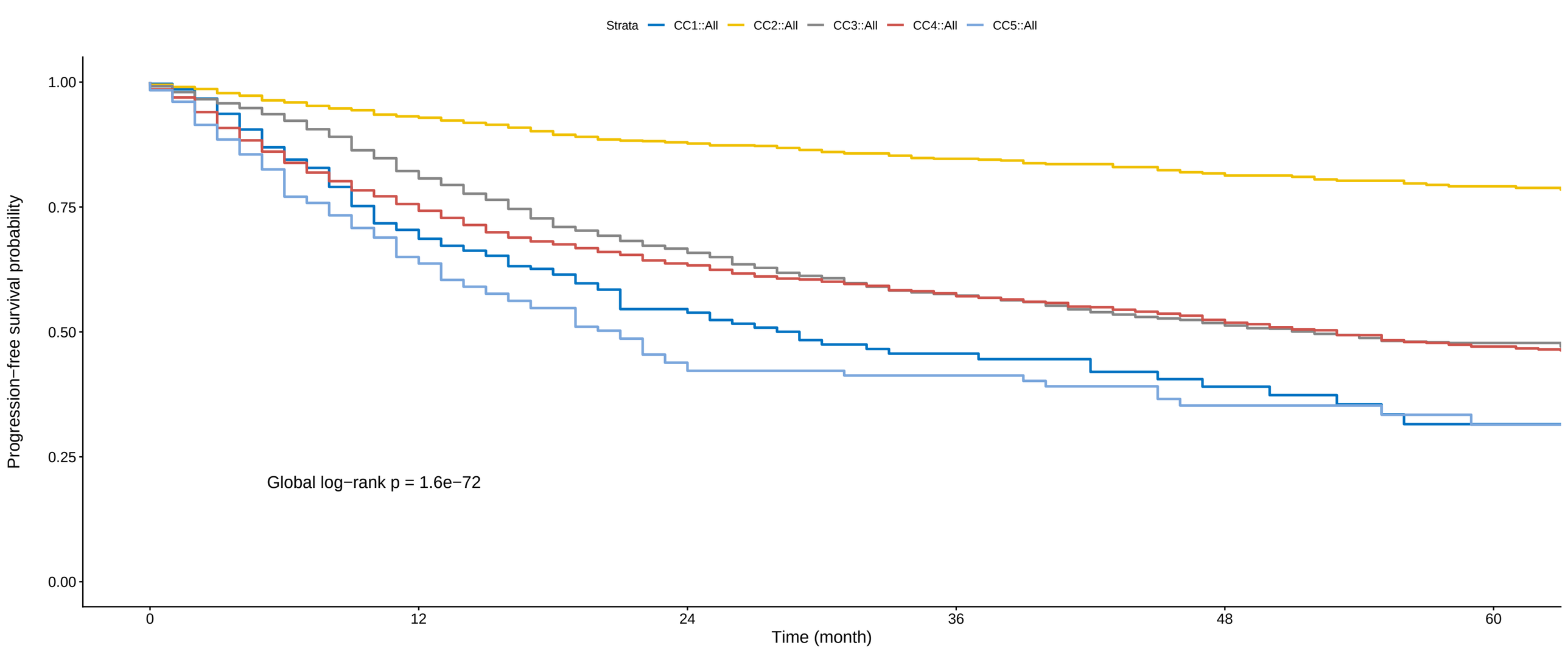

Cell−Cycle CCD by Cluster
CCD
0.1
0.0
−0.1
−0.2
4
2
5
3
1
LIHC.iCluster:1
LIHC.iCluster:2
LIHC.iCluster:3
HNSC.Atypical
BLCA.1
PRAD.8−other
HNSC
COAD
PRAD.5−SPOP
PRAD.1−ERG
KIRC
KIRC.2
UCEC
KICH.Eosin.0
KIRP.C2a
THCA.2
THCA
KIRC.4
KIRP
KIRC.1
KIRP.C1
THCA.3
THCA.5
STAD
LUSC
THCA.4
THCA.1
UCEC.MSI
UCEC.CN_LOW
UCEC.NA
UCEC.POLE
STAD.GI.EBV
COAD.GI.GS
SKCM.BRAF_Hotspot_Mutants
OVCA.Proliferative
STAD.GI.CIN
SKCM.Triple_WT
SKCM.RAS_Hotspot_Mutants
OVCA.Mesenchymal
STAD.GI.MSI
READ.GI.CIN
COAD.GI.MSI
HNSC.Classical
COAD.GI.CIN
LUSC.basal
OVCA.Immunoreactive
UCEC.CN_HIGH
BRCA.LumB
OVCA.Differentiated
BRCA.Basal
BRCA.Her2
BLCA.3
LUSC.classical
UCS.2
LGG.GBM.LGG.Mesenchymal.like
STAD.GI.GS
BRCA
LIHC
LUAD
GBM.GBM.LGG.Classic.like
AML.5
LUAD.4
LGG.GBM.LGG.G.CIMP.high
HNSC.Basal
BRCA.Normal
HNSC.Mesenchymal
LUAD.5
ACC.CIMP−low
LGG.GBM.LGG.Codel
BRCA.LumA
GBM.GBM.LGG.NA
GBM.GBM.LGG.Mesenchymal.like
KIRC.3
PRAD
BLCA.2
AML.6
LUAD.6
AML.4
LUAD.2
LUSC.secretory
LUAD.3


Both−Oscillator CCD by Cluster
CCD
−0.10
−0.15
−0.20
4
2
5
3
1
PRAD
STAD.GI.GS
KIRC.3
BRCA
LGG.GBM.LGG.Mesenchymal.like
LUAD
LIHC
KIRP
COAD
THCA.4
KIRC.4
PRAD.5−SPOP
STAD
THCA.3
KIRC.2
LIHC.iCluster:1
UCEC
LUSC
PRAD.1−ERG
THCA.2
THCA
THCA.1
PRAD.8−other
KIRC
LIHC.iCluster:2
KIRP.C2a
LIHC.iCluster:3
KIRP.C1
THCA.5
KICH.Eosin.0
KIRC.1
HNSC
HNSC.Atypical
BLCA.1
STAD.GI.EBV
COAD.GI.MSI
READ.GI.CIN
STAD.GI.CIN
COAD.GI.CIN
UCEC.CN_LOW
STAD.GI.MSI
OVCA.Mesenchymal
LUSC.basal
SKCM.BRAF_Hotspot_Mutants
OVCA.Differentiated
UCEC.POLE
SKCM.RAS_Hotspot_Mutants
UCEC.MSI
BLCA.3
BRCA.LumB
HNSC.Classical
BRCA.Basal
BRCA.Her2
UCEC.2
OVCA.Immunoreactive
SKCM.Triple_WT
UCEC.CN_HIGH
COAD.GI.GS
OVCA.Proliferative
UCEC.NA
LUSC.classical
LUAD.5
LUAD.6
LUAD.3
BRCA.Normal
LUAD.4
ACC.CIMP−low
LUSC.secretory
BRCA.LumA
BLCA.2
HNSC.Mesenchymal
LGG.GBM.LGG.G.CIMP.high
LUAD.2
LGG.GBM.LGG.Codel
AML.5
GBM.GBM.LGG.NA
GBM.GBM.LGG.Classic.like
AML.6
GBM.GBM.LGG.Mesenchymal.like
HNSC.Basal
AML.4



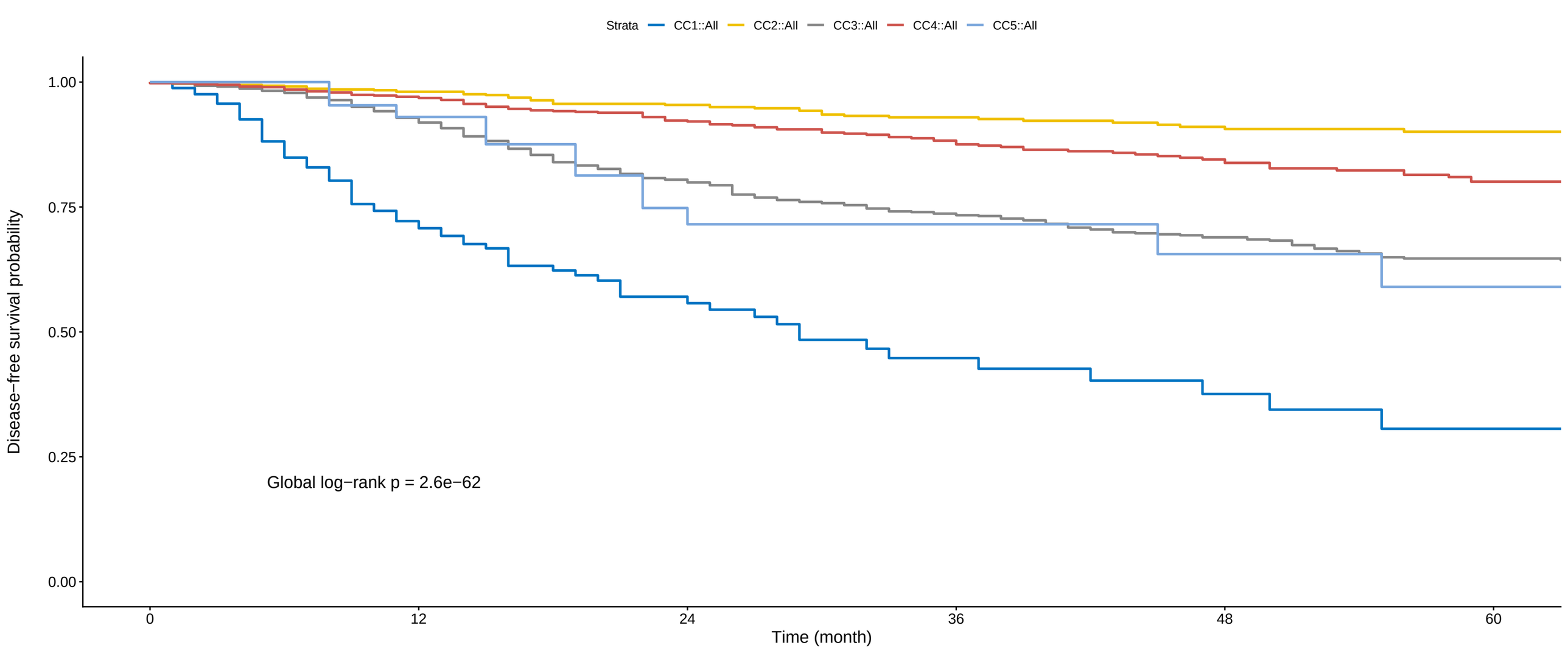
Number at risk

CC1::All	291	206	106	67	46	27
CC2::All	1243	1096	819	579	418	304
CC3::All	2746	2163	1448	984	711	513
CC4::All	2201	1609	986	683	454	331
CC5::All	182	128	81	57	36	23



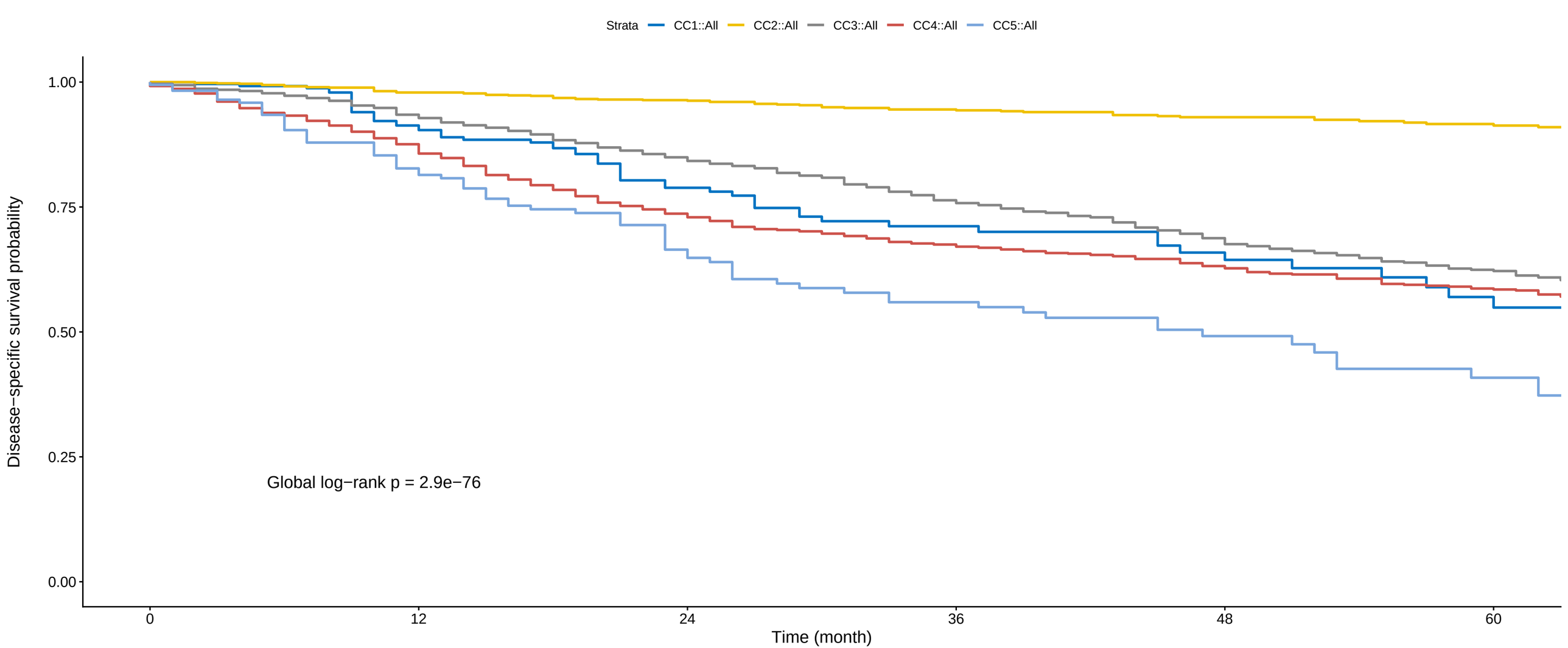
Number at risk

CC1::All	291	158	75	43	24	12
CC2::All	1240	1040	743	518	366	261
CC3::All	2748	1892	1117	701	500	353
CC4::All	2100	1345	814	563	367	254
CC5::All	183	100	54	42	24	16



Number at risk

CC1::All	177	102	44	22	13	6
CC2::All	690	629	449	291	204	153
CC3::All	1503	1211	731	465	329	231
CC4::All	915	767	501	357	244	172
CC5::All	46	40	23	17	11	8



Number at risk

CC1::All	280	199	103	66	45	27
CC2::All	1228	1083	809	573	412	299
CC3::All	2637	2091	1398	947	690	501
CC4::All	2011	1501	915	633	424	308
CC5::All	180	127	81	57	36	23